UNIVERSITY OF MANCHESTER

STRUCTURAL GEOLOGY

Date: whatever

Time: whatever

Duration: 1 hour

In an area of large-scale folding the following data was collected from around an **antiform**:

bedding: 056/83NW, 058/86NW, 061/87SE, 064/82SE, 060/V, 071/70S, 085/51S, 108/36S, 130/30SW, 008/34W, 011/35W, 014/36W, 018/38W, 022/40W

minor fold axes: 26/240, 28/242, 30/239

trend of a cleavage which is axial planar to the minor folds: 036°

By plotting a stereogram fully describe the geometry of the fold and draw a simplified profile plane section to illustrate that geometry. Marks will be awarded on the following basis:

Stereoprojection (/30)

(a) Intelligibility:
- Sensible title **[1]**
- Key present and comprehensive **[2]**
- Terms in key fully specified **[1]**
- Neatness **[1]**

(b) Accuracy of plotting:
- Accurately plotted poles to bedding **[2]**
- Accurately plotted minor fold axes **[2]**
- Accurately plotted trace of cleavage **[2]**

(c) Well chosen values for the following:
- Fold limbs (shown as great circles) **[2]**
- Axial plane (shown as great circle) **[1]**
- Bisector (shown as great circle) **[1]**
- Fold axis **[1]**
- Interlimb angle **[1]**

(d) Correctly specified orientations giving:
- Representative limbs **[2+2]**
- Fold axis **[2]**
- Axial plane **[2]**
- Bisector **[2]**
- Interlimb angle **[2]**
- Quoted fold axis is exactly the pole to the quoted profile plane **[1]**

Simplified profile plane section (/20)

(e) Intelligibility of presentation:
- An unambiguously worded title incorporating an indication of whether view is up- or down-plunge, and the strike/dip of the plane of section **[3]**
- Correctly positioned orientation indicators **[1]**
- Neatness **[1]**

(f) Intelligibility of content:
- Labelling of limbs **[1]**
- Correctly located axial plane **[1]**
- Labelling of axial plane **[1]**
- Labelling of bisector **[1]**
- Labelling of interlimb angle **[1]**

(g) Accuracy of apparent dips (use of true dips = 0):
- Correctly drawn as antiform or synform **[2]**
- Apparent dips in the right orientation **[1]**
- All apparent dips labelled appropriately **[1]**
- Accurate apparent dip of limbs **[2]**
- Accurate apparent dip of axial plane **[1]**
- Accurate apparent dip of bisector **[1]**
- Apparent dips consistent (to within 1°) with interlimb angle **[1]**
- Relative thickness of the limbs shown **[1]**